MIL-D-48792(PA)
27 January 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

DELAY ELEMENT, ADAPTION KIT: M40

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of relay entitled Delay Element, Adaption Kit: M40 which is used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8840143 - Delay Element.
EL-8854182 - Equipment List.

FSC:1135

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The delay element shall comply with all requirements specified on the drawing (dwg) 8840134, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Timing. The delay elements shall be energized using the circuit shown on Figure 1. The circuit between terminal pins W and F of J1 shall close within 30 $\pm$ 13 seconds after the closure of switch S1. At temperature extremes (-65 degrees Fahrenheit (°F) and 160°F). The circuit shall close within 30 $\pm$ 15 seconds. Upon closure of this circuit the voltage between J1 -E and J1 -K shall be 105 volts direct current (vdc) minimum, terminal J1 -E shall be at a positive potential with respect to terminal J1 -K. The circuit between terminals W and F of J1 shall not close prior to 14 + 1 seconds after the 50 vdc is applied. The test shall be as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

From Terminal	To Terminal	Timing Requirement (seconds)	Test Code Nos.
J1-W	J1-F	30 ± 13	030001

From Terminal	To Terminal	Voltage Requirement (VDC)	Test Code Nos.
J1-E	J1-K	105 min	030002

3.7 Circuit resistance. The circuit resistance shall be as specified below when tested as specified in 4.4.2. For tests number 7 through 12, the relays are to be actuated either by using the circuit shown in Figure 1 or by placing 115 ± 5 vdc between terminal J1-E and J1-K.

Test No.	Between Terminals		Resistance (ohms)	
1	J1-P	J2-G	0.1 max	Relays non-actuated
2	J1-c	J2-P	0.1 max	Relays non-actuated
3	J1-$\overline{d}$	J2-N	0.1 max	Relays non-actuated
4	J1-$\overline{R}$	J2-F	0.1 max	Relays non-actuated
5	J1-S	J2-L	0.1 max	Relays non-actuated
*6	J1-E	J1-K	6000 ± 525	Relays non-actuated
7	J1-M	J1-L	0.1 max	Relays actuated
8	J1-N	J1-b	0.1 max	Relays actuated
9	J1-J	J1-$\overline{Y}$	0.1 max	Relays actuated
10	J1-H	J1-Z	0.1 max	Relays actuated
11	J1-G	J1-X	0.1 max	Relays actuated
12	J1-F	J1-W	0.1 max	Relays actuated

*Note: This test is to be made only at room temperature.

From Terminal	To Terminal	Resistance (ohms)	Test Code Nos.
J1-E	J1-K	6000 ± 525	040001
J1-M	J1-L	0.1 max	040002
J1-N	J1-b	0.1 max	040003
J1-J	J1-$\overline{Y}$	0.1 max	040004
J1-H	J1-Z	0.1 max	040005
J1-G	J1-X	0.1 max	040006
J1-F	J1-W	0.1 max	040007

3.8 Environmental testing. The delay element shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The delay element shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

3.8.1 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the terminals and all other terminals listed below, when tested as specified in 4.4.3.1. All unused terminals shall be connected together and considered as one terminal.

Receptacle J1 - F,G,H,J,L,N,P,R,S,W,X,Y,Z,b,c,d.

Unused terminals

Receptacle J1 - A,B,T,U,V,a,e,f,g.

Receptacle J2 - B,E,J,N,R,S,T,U,V,W,X,Y,Z.

3.8.2 Low temperature. The delay element shall show no evidence of damage or deterioration upon exposure to temperatures of -65 ± 5°F and while at this temperature shall meet the requirements of timing (3.6) and insulation resistance (3.8.1), when tested as specified in 4.4.3.2. After completion of the test, the delay element shall meet the requirements of circuit resistance (3.7) at 80 ± 20°F.

3.8.3 High temperature. The delay element shall show no evidence of damage or deterioration upon exposure to temperatures of 160 ± 5°F and while at this temperature, the assembly shall meet the requirements of timing (3.6) and insulation resistance (3.8.1), when tested as specified in 4.4.3.3. After completion of the test the delay element shall meet the requirements of circuit resistance (3.7) at 80 ± 20°F.

3.8.4 Humidity. The delay element shall withstand exposure to humidity without evidence of damage affecting function when tested as specified in 4.4.3.4. Within one hour after completion of the test, the delay element shall meet the requirements of timing (3.6), circuit resistance (3.7), and insulation resistance (3.8.1).

3.8.5 Shock. The delay element shall show no evidence of damage when subjected to shocks of 30 ± 5 g's, three times in each direction along three mutually perpendicualr axes for a total of 18 shocks, when tested as specified in 4.4.3.5. After completion of the test the delay element shall meet the requirements of timing (3.6), circuit resistance (3.7), and insulation resistance (3.8.1).

3.8.6 Transportation vibration. The delay element shall withstand vibration for 15 minutes in each of three mutually perpendicular axes at the frequencies and amplitudes specified below, when tested as specified in 4.4.3.6. The frequency shall be varied uniformly from 8 to 500 to 8 Hertz (Hz) every 15 minutes. Following this test the delay element shall meet the requirements of timing (3.6), circuit resistance (3.7) and insulation resistance (3.8.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Frequency Range (Hz)	Cycle Time	Constant Acceleration or Double Amplitude
8-55-8	7.5 min	0.060 ± 0.006 inch
55-500-55	7.5 min	10 ± 1 g

3.9 Test codes. The contractor shall provide data feed back on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8840134) for first

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspections and tests

Inspection or Test	Testing Sequence of Unit No. 1	2	3	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)	1	1	1		X
2. Insulation resistance (see 3.8.1)	2	2	2		X
3. Timing requirements (see 3.6)	3	3	3		X
4. Circuit resistance (see 3.7)	4	4	4		X
5. Transportation vibration (see 3.8.6)	5	6	9		X
6. Low temperature (see 3.8.2)	8	7	6		X
7. High temperature (see 3.8.3)	7	5	7		X
8. Shock (see 3.8.5)	6	8	8	X	
9. Humidity (see 3.8.4)	9	9	5	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

Source: https://assist.dla.mil- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Delay element (prior to assembly in case, (see dwg 8840134).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Components missing	Visual
	102. Improper mounting or location of components	Visual
	103. Improper soldering	Visual
	104. Damaged parts	Visual
	105. Foreign material	Visual
	106. Damaged insulation	Visual
	107. Loose components	Visual
	108. Incorrect assembly	Visual
	109. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Incorrect or illegible markings	Visual
	202. Incorrect wire size	Visual
	203. Incorrect wire type	Visual
	204. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 Delay element (prior to potting, see dwg 8840134).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Major:
101. Connectors improperly oriented.....Visual
102. Loose connectors.....Manual

Minor: None defined.

4.3.2.1.3 Delay element, M40 (see dwg 8840134).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper location of mounting holes	Gage
102.	Improper size of mounting hole	Gage
103.	Connectors will not mate	Gage
104.	Damaged connector	Visual
105.	Incorrect potting material	Visual
*106.	Incorrect height of pins	Visual
**107.	Incorrect depth of sockets	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Improper potting level	Visual
202.	Nameplate incorrect or illegible	Visual
203.	Nameplate missing	Visual

*In the event pins do not appear to be within 1/32 inch in height relative to the adjacent pins, gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch max from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements.

**Same as above except socket insert depths shall be substituted for pin heights.

4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging, packing and marking in accordance with dwg 8840134	Visual

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

Minor:
201. Workmanship (see 3.11).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Timing	Major
b. Circuit resistance	Major
c. Insulation resistance	Major
d. High temperature	Major
e. Low temperature	Major
f. Transportation vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Timing, circuit resistance and insulation resistance. Each delay element shall be tested in accordance with 4.4.1 through 4.4.3.1. If any item fails to comply with 3.6, 3.7 and 3.8.1, the item shall be rejected and removed from the lot.

4.3.3.2 Low temperature, high temperature and transportation vibration. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.3.2, 4.4.3.3 and 4.4.3.6. If any item fails to comply with 3.8.2, 3.8.3 and 3.8.6, the lot shall be rejected.

Table IV

Sample sizes for non-destructive testing

Lot Size	Sample Size
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854182). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Timing. Connect the circuit shown in Figure 1 to the assembly. The 400 Hertz supply used in testing shall be isolated from power line ground. Close switch S1 and measure the time between the switch closure and closure of the circuit between terminals W and F of the J1 receptacle. Upon closure of this circuit measure the voltage between terminals J1-E and J1-K using a dc voltmeter with an input impedance of 1000 ohm per volt minimum and with a maximum instrument error of 3 percent. To test for non-closure of the circuit upon removal of supply voltage, switch S1 shall be closed for the specified time and then opened. There shall be no indication of circuit closure in excess of 20 microseconds between J1-W and J1-F when switch S1 is opened. There shall be a minimum time of one minute between the test for delay time and the test for non-closure with no voltage applied to terminal J1-C. Record delay time and voltage.

4.4.2 Circuit resistance. The circuit resistance of the delay element shall be measured between the terminals listed in 3.7 using a test current limited to 5 milliamperes (ma) or less. Record resistances for tests No. 6 through 12.

4.4.3 Environmental testing (see 3.8).

4.4.3.1 Insulation resistance. The insulation resistance of the delay element shall be measured between terminals listed in 3.8.1 using a test voltage of 500 ± 50 volts with current limited to 5 ma. All unused terminals may be connected together and considered as one terminal.

4.4.3.2 Low temperature. The delay element shall be placed in a test chamber. The temperature within the chamber shall be lowered to -65 ± 5°F and held at this temperature for a minimum of 4 hours after the temperature has stabilized (see 6.4). After 4 hours and while still at -65 ± 5°F, the assembly shall be tested for compliance with the requirements of timing (3.6) and insulation resistance (3.8.1). After the

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

temperature of the assembly has returned to 80 ± 5°F, the assembly shall be tested for compliance with the requirement for circuit resistance (3.7).

4.4.3.3 High temperature. The delay element shall be placed in a test chamber. The temperature within the chamber shall be raised to +160 ± 5°F for a minimum of 4 hours after the temperature has stabilized (see 6.4). After 4 hours and while still at 160 ± 5°f, the assembly shall be tested for compliance with the requirements of timing (3.6) and insulation resistance (3.8.1). After the temperature of the assembly shall be tested for compliance with the requirement for circuit resistance (3.7).

4.4.3.4 Humidity. The delay element shall be placed in a humidity test chamber and shall be subjected to the humidity test as specified in MIL-E-5272, paragraph 4.4, procedure 1 except that the number of cycles shall be three. All temperatures shall be maintained at ± 5°F. The ph value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the last cycle, the delay element shall be removed from the test chamber and allowed to return to ambient temperature (80 ± 20°F) without forced cooling. After returning to ambient temperature, the delay element may be wiped and then blown with filtered compressed air for a period of 5 minutes and then shall be tested for compliance with the requirements of timing (3.6), circuit resistance (3.7), and insulation resistance (3.8.1).

4.4.3.5 Shock. The delay element shall be subjected to a total of eighteen shocks of 30 ± 5 g's, three in each direction along each of three mutually perpendicular axes. The shock axes shall conform to the axes through the centers of the two connectors and to the other two axes mutually perpendicular to this longitudinal axis and each other. The time duration of each shock shall be 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of the peak value. The time to peak shall be 5.5 ± 1 ms. After the test, the assembly shall be tested for compliance with the requirements of timing (3.6), circuit resistance (3.7), and insulation resistance (3.8.1).

4.4.3.6 Transportation vibration. The delay element shall be mounted on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes (4.4.3.5) at the frequencies and amplitude specified in 3.8.6. The total test time shall be 45 minutes. After the test, the assembly shall be tested for compliance with the requirements of timing (3.6), circuit resistance (3.7), and insulation resistance 3.8.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The delay element shall be packaged in accordance with dwg 8840134 using the specified moisture barrier.

5.1.2 Level C. The delay element shall be packaged in accordance with dwg 8840134 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The delay element shall be packed in accordance with dwg 8840134.

5.3 Marking. Marking shall be as specified in accordance with dwg 8840134.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 3.5 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2144 Rev.1 dated 15 January 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A174

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:40Z
Check the source to verify that this is the current version before use.